Advanced Full Stack Web Development

https://tutors.dev/course/full-stack-oth-25

SETU
Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

OTH
OSTBAYERISCHE TECHNISCHE HOCHSCHULE REGENSBURG

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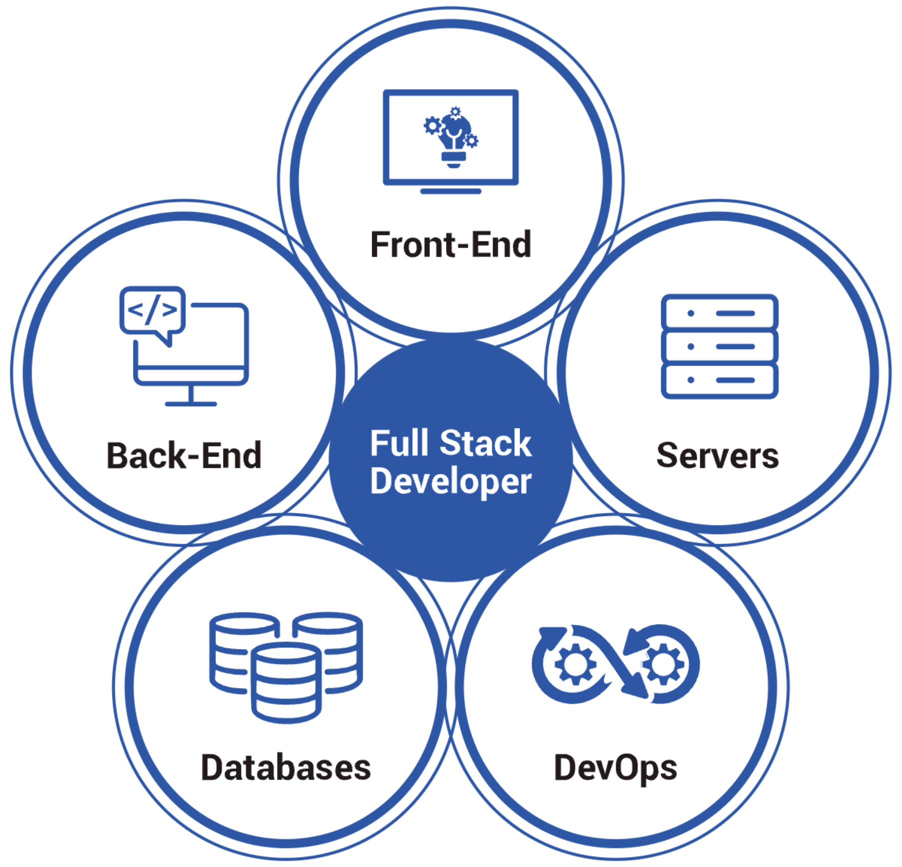
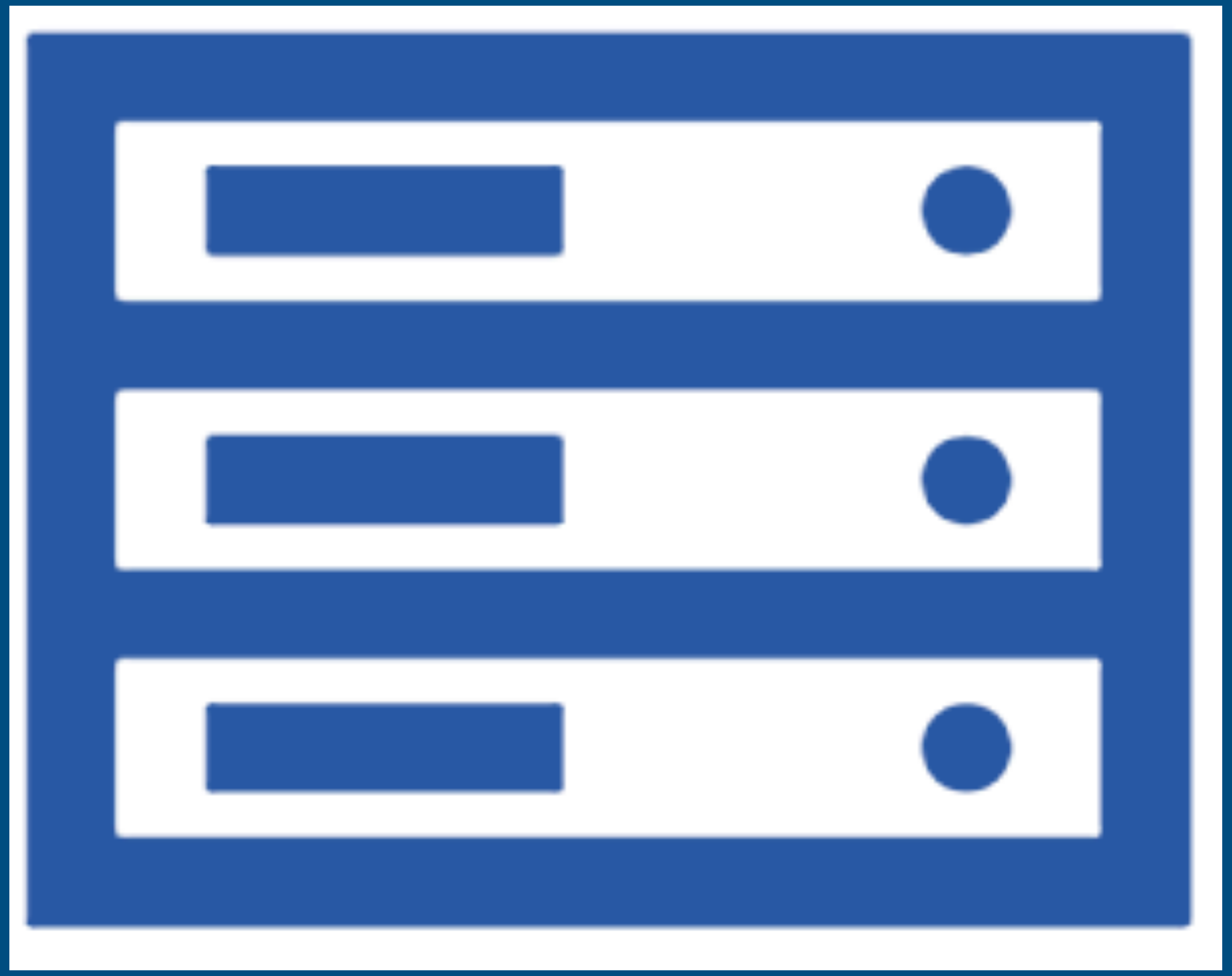
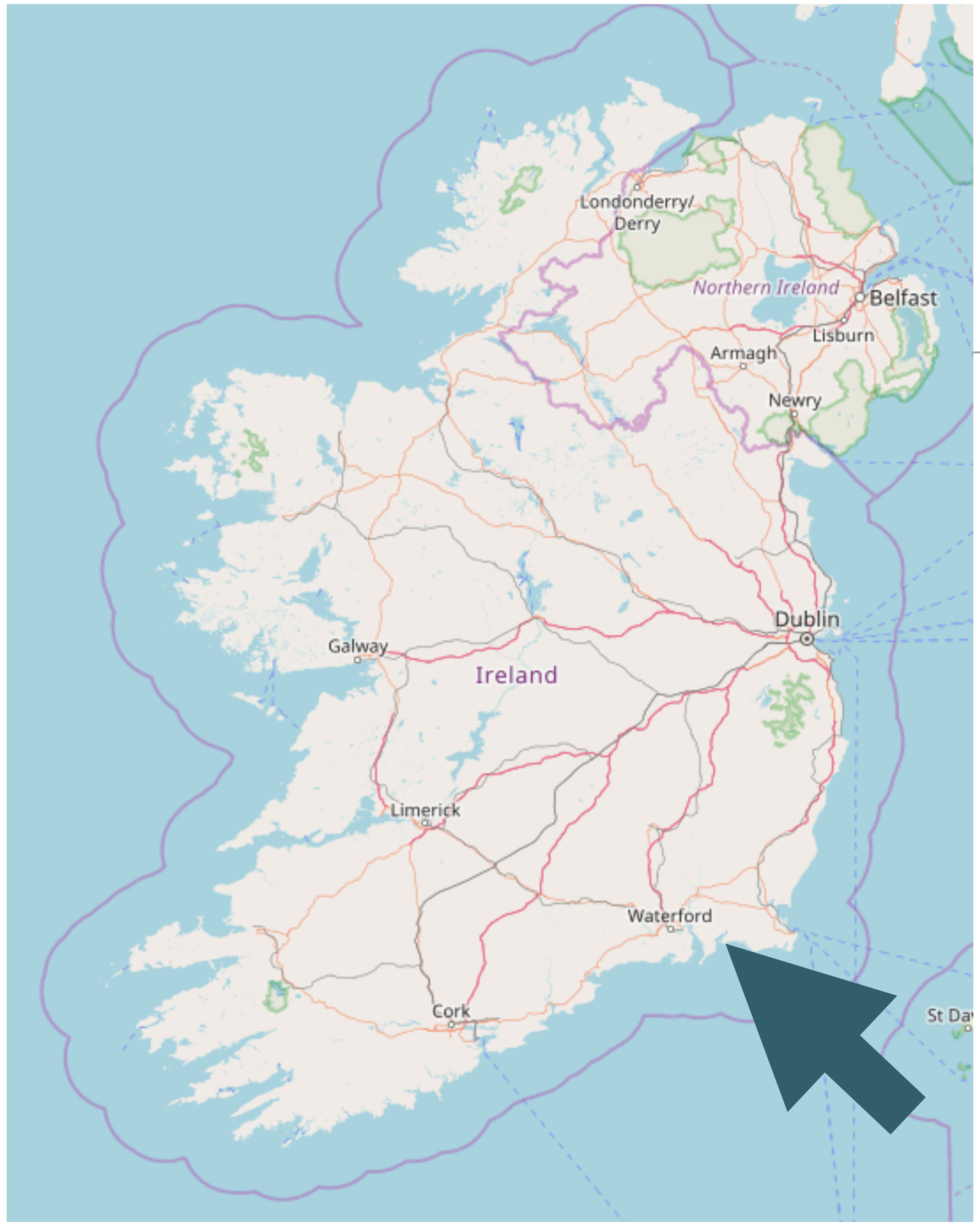


Course Overview

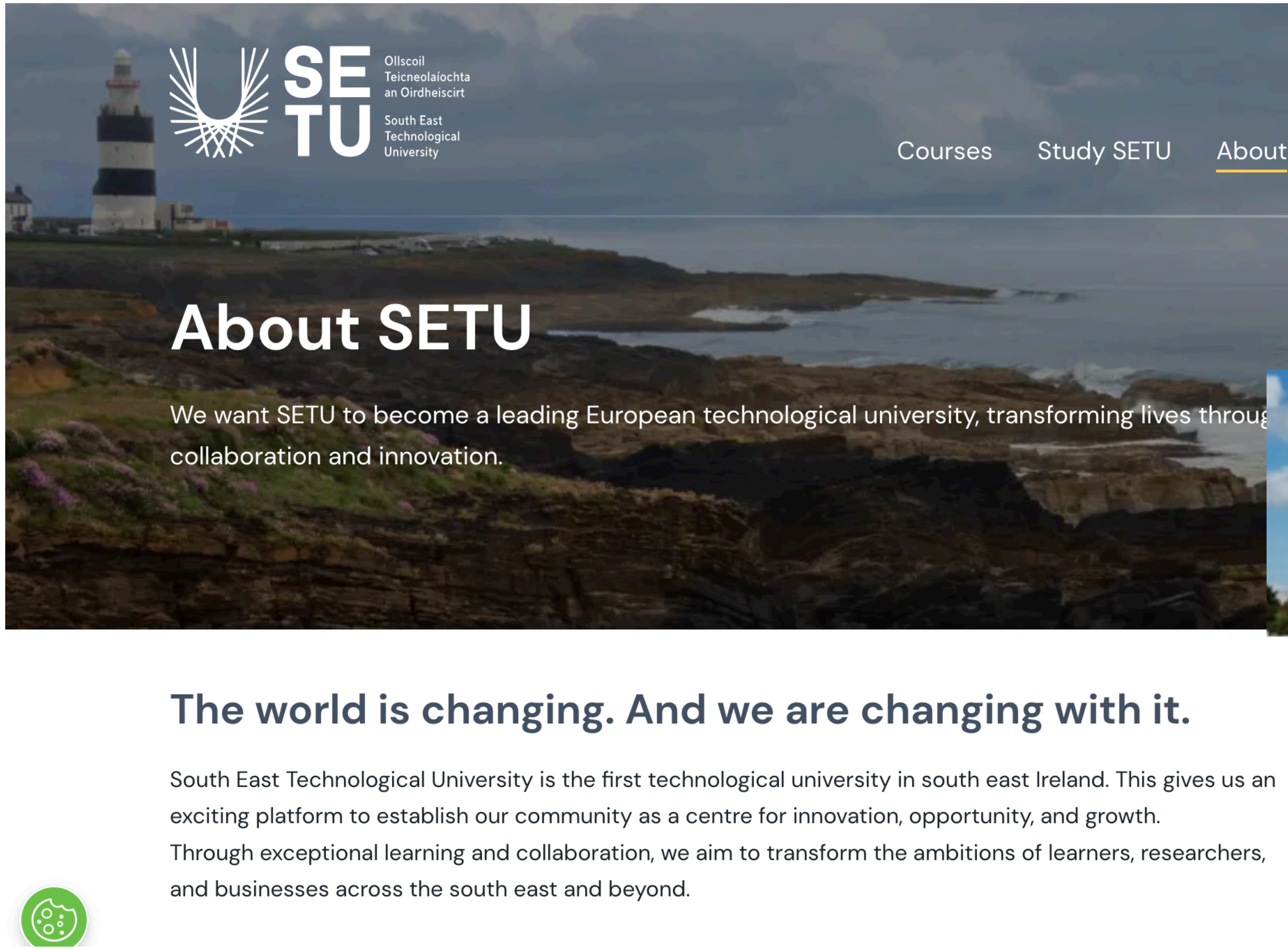


Advanced Stack Web Development

Waterford



Waterford (from Old Norse *Veðrafjǫrðr*, meaning "ram (wether) fjord", Irish: *Port Láirge*) is a city in Ireland. It is in the South-East Region, Ireland and is part of the province of Munster. The city is situated at the head of Waterford Harbour. It is the oldest[2][3] and the fifth most populous city in the Republic of Ireland. It is the eighth most populous city on the island of Ireland.



South East Technological University is a university-level institution in the South-East of Ireland with over 18,000 students and 1,700 staff.

SETU offers tuition and research programmes in various areas from Higher Certificate to Degree to PhD.

Agenda

- Prerequisites
- Module Overview
- Components
- References

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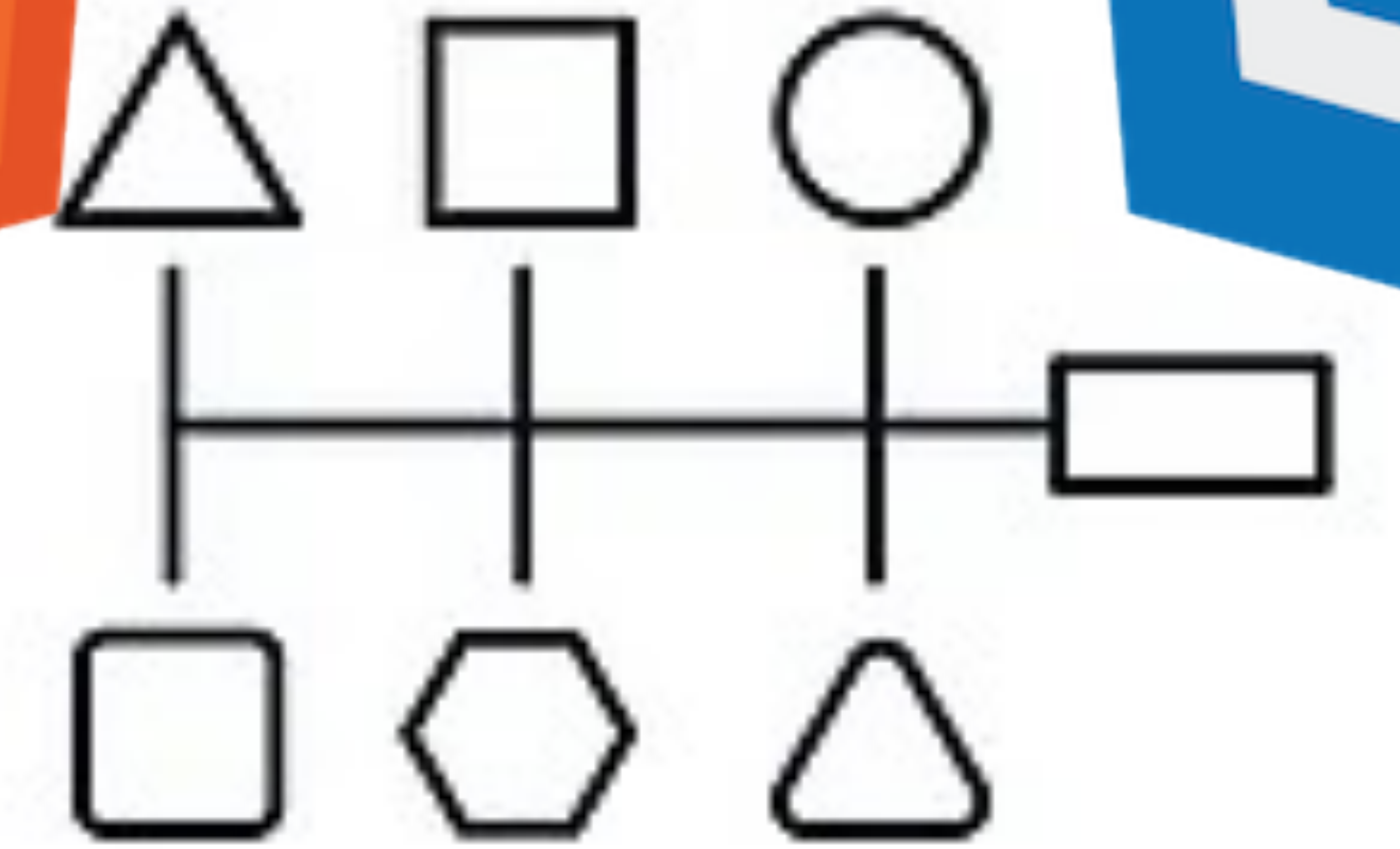
Perquisites

- Foundation level skills in:

  - **HTML**: Ability to effectively structure the html content of a small to medium static site, including some understanding of the use of templates

  - **CSS:** Understand the fundamentals of CSS, and be able to realise simple layouts and designs. Be comfortably using a CSS framework



- **Programming**: Intermediate level programming programming ability in Javascript or any C like language (Java, C#, Go etc)

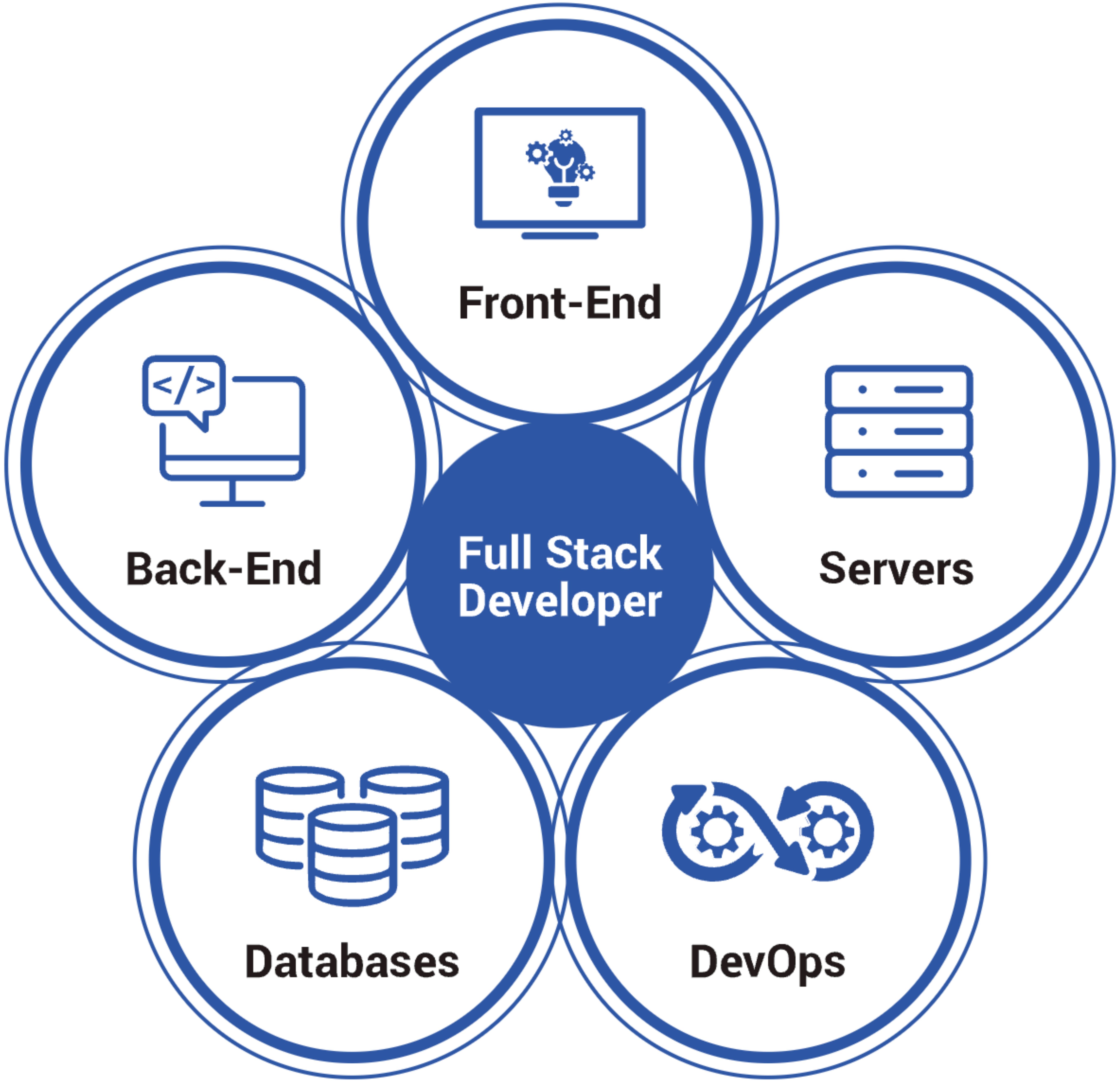
Back End Web Development

Test Driven Development

Application Programmer Interfaces

Front End Web Development

Full Stack



Core Concepts

Back End Web Development

Test Driven Development

Application Programmer Interfaces

Front End Web Development

Full Stack

01: Node

The fundamentals of Node. A first Node application.

02: Hapi.js

The Hapi application Framework

03: Joi

Validation with Joi

05: Models

Incorporate Mongo Models

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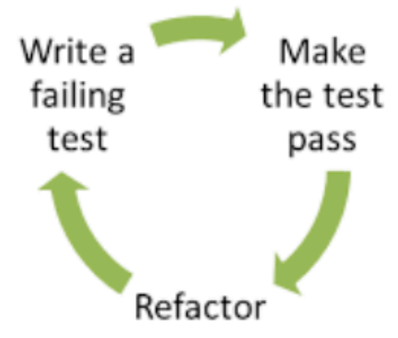
04: TDD

Test Driven Development

TDD Introduction



The fundamentals of Test Driven Development

Right BICEP

Guidelines for Tests · Right · Boundary · Inverse · Cross-check · Errors ·...

C.O.R.R.E.C.T

More TDD guidelines · Conformance · Ordering · Range · Reference · Existin...

FIRST Principles

More TDD Guidelines: Fast · Isolate · Repeatable · Self-Validating · Timely

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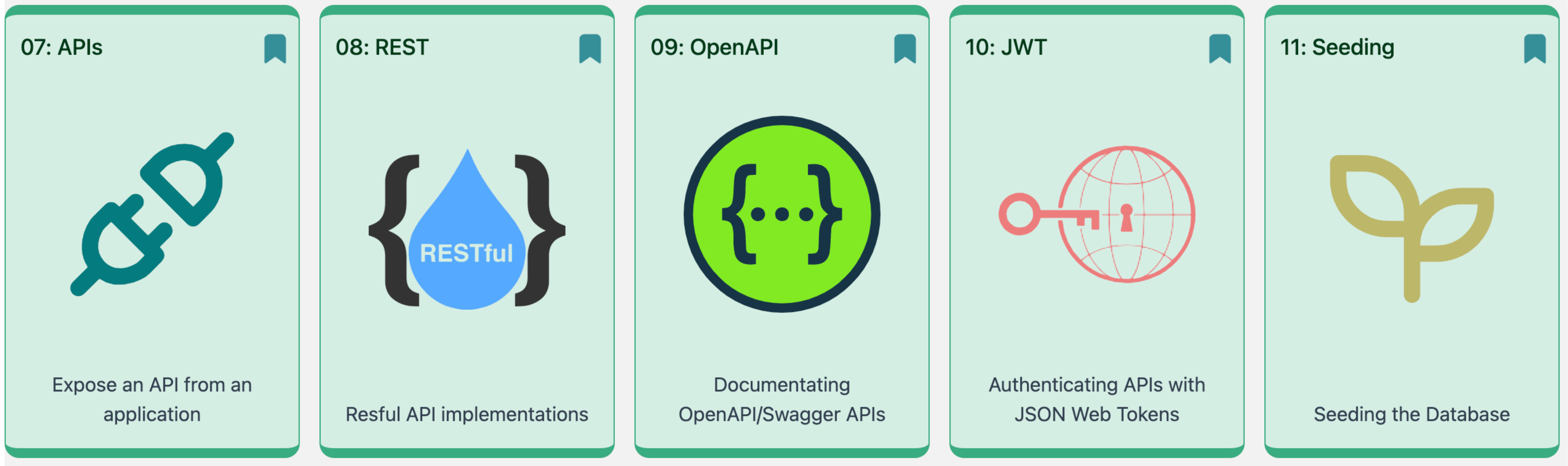

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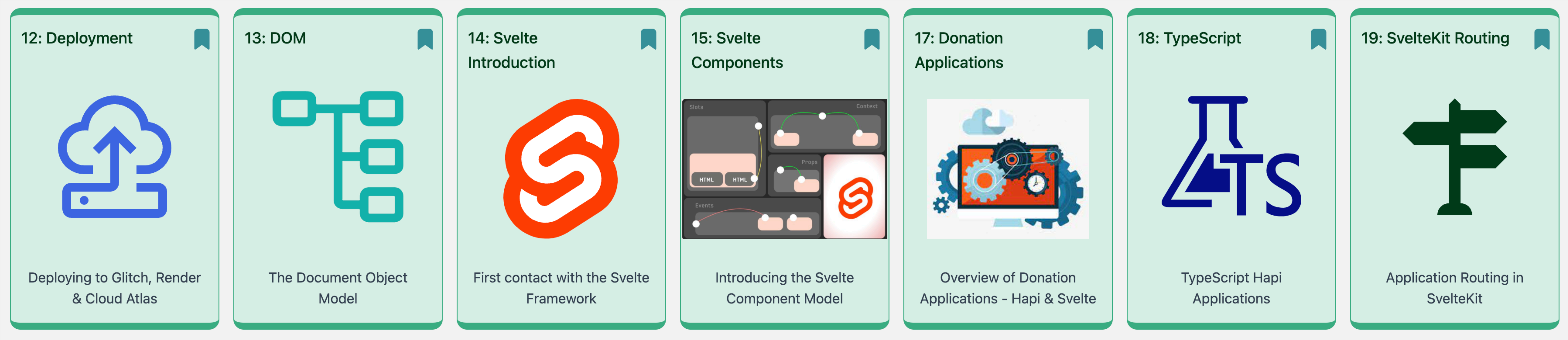

Back End Web Development

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Application Programmer Interfaces

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Full Stack

20: Svelte Applications

The Structure of Applications in Svelte

21: Charts & Graphs

Frappe Charts and Graphs in Svelte

22: Maps

Mapping components in Svelte

23: Server Side Rendering

SSR patterns in SvelteKit

24: Full Stack Development

Patterns for Full Stack in SvelteKit

- **01: Node** — The fundamentals of Node. A first Node application.
- **02: Hapi.js** — The Hapi application Framework
- **03: Joi** — Validation with Joi
- **05: Models** — Incorporate Mongo Models

- **04: TDD** — Test Driven Development
- **TDD Introduction** — The fundamentals of Test Driven Development
- **Right BICEP** — Guidelines for Tests · Right · Boundary · Inverse · Cross-check · Errors ·...
- **C.O.R.R.E.C.T** — More TDD guidelines · Conformance · Ordering · Range · Reference · Existin...
- **FIRST Principles** — More TDD Guidelines: Fast · Isolate · Repeatable · Self-Validating · Timely
- **07: APIs** — Expose an API from an application
- **08: REST** — Resful API implementations
- **09: OpenAPI** — Documentating OpenAPI/Swagger APIs
- **10: JWT** — Authenticating APIs with JSON Web Tokens
- **11: Seeding** — Seeding the Database

- **12: Deployment** — Deploying to Glitch, Render & Cloud Atlas
- **13: DOM** — The Document Object Model
- **14: Svelte Introduction** — First contact with the Svelte Framework
- **15: Svelte Components** — Introducing the Svelte Component Model
- **17: Donation Applications** — Overview of Donation Applications - Hapi & Svelte
- **18: TypeScript** — TypeScript Hapi Applications
- **19: SvelteKit Routing** — Application Routing in SvelteKit

- **20: Svelte Applications** — The Structure of Applications in Svelte
- **21: Charts & Graphs** — Frappe Charts and Graphs in Svelte
- **22: Maps** — Mapping components in Svelte
- **23: Server Side Rendering** — SSR patterns in SvelteKit
- **24: Full Stack Development** — Patterns for Full Stack in SvelteKit

Assessment

Students will be given a comprehensive specification of an application - including a detailed grading rubric

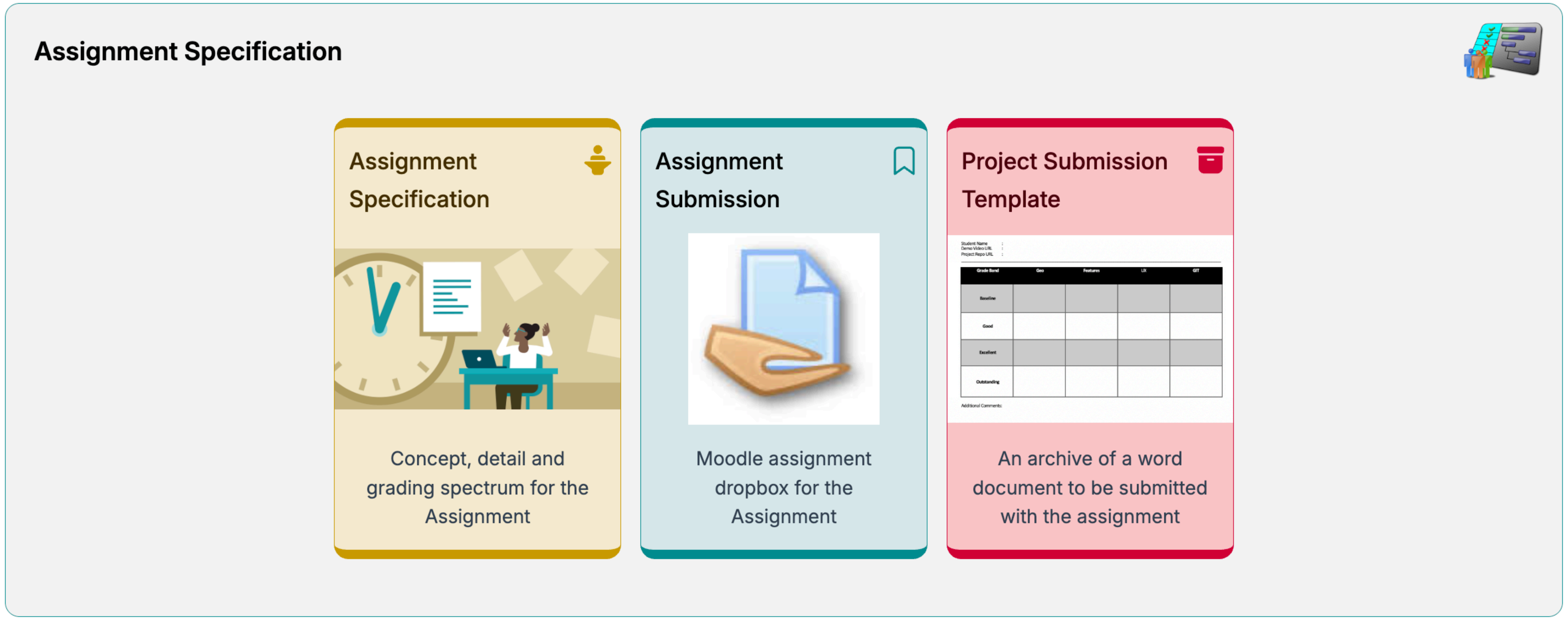
The projects will be submitted on GitHub, and will be a suitable candidate for the students personal portfolio

Students will prepare a demonstration video walking through the assignment in detail + an interview (Zoom)

Delivery Modes

Onsite classes in Regensburg - October 21-30, 2025

Virtual classes will be via recorded videos + laboratories + a supervised Slack channel



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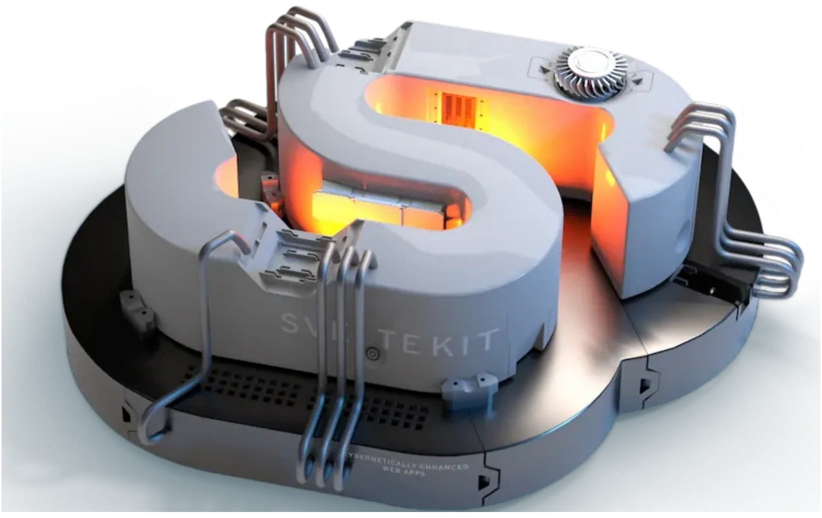
Onsite schedule October 21-30, 2025

	Tuesday 21	Wednesday 22	Thursday 23	Tuesday 28	Wednesday 29	Thursday 30
8:15-9:45	K139			K139		
10:00-11:30						
11:45-13:15			K139			K139
13:45-15:15		K139	K139		K139	K139
15:30-17:00		K139	K139		K139	K139
17:15-18:45		K139			K139	



hapi

JS

mongoDB. Atlas

MOCHA

node JS

OpenWeather

joi

WebStorm

chai

TypeScript

JavaScript that scales.

JUnit

GitHub Developer

SVELTE

Swagger

Visual Studio Code

Firebase

Leaflet

netlify

JWT

mongoDB

mongoose

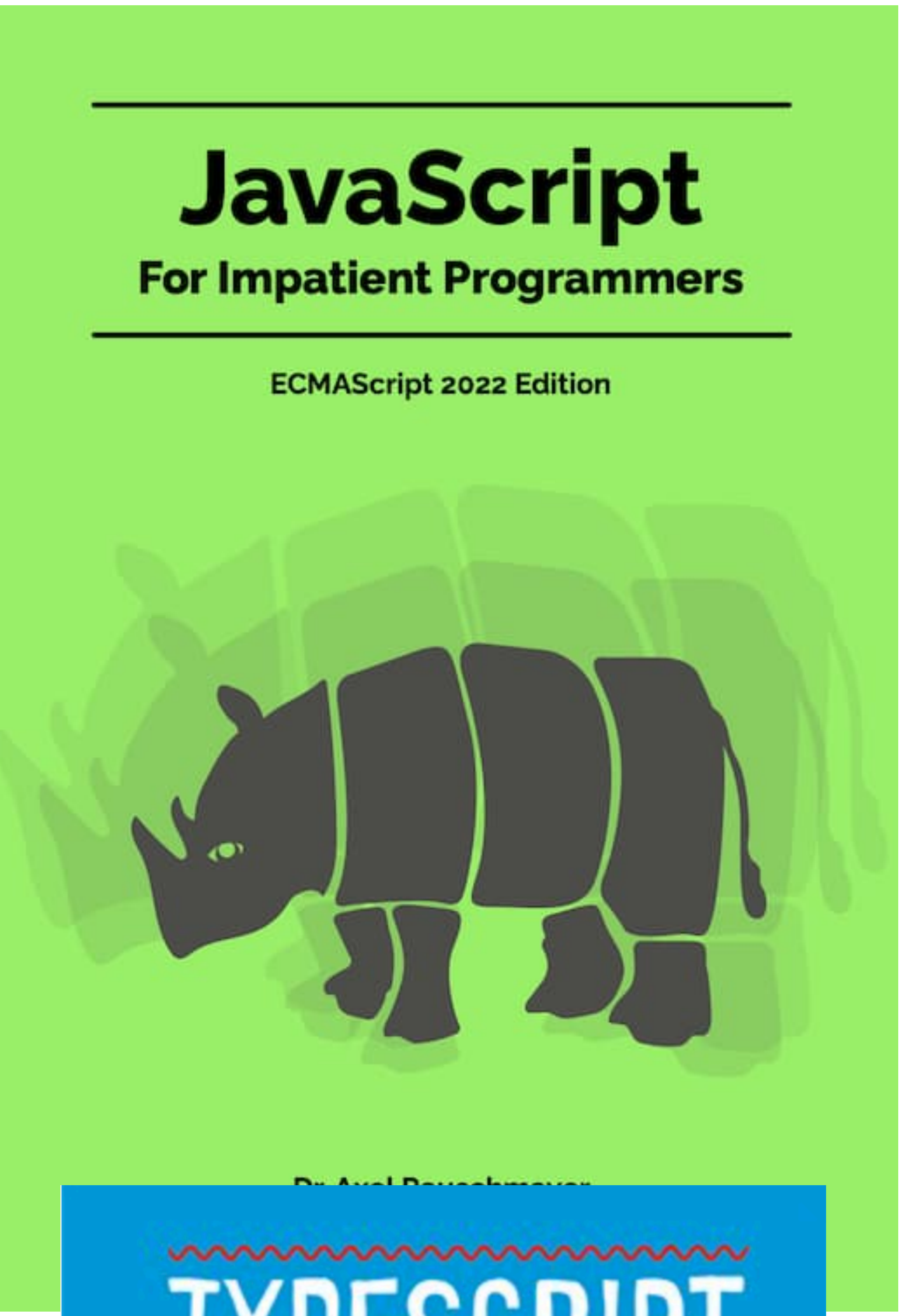
Vercel

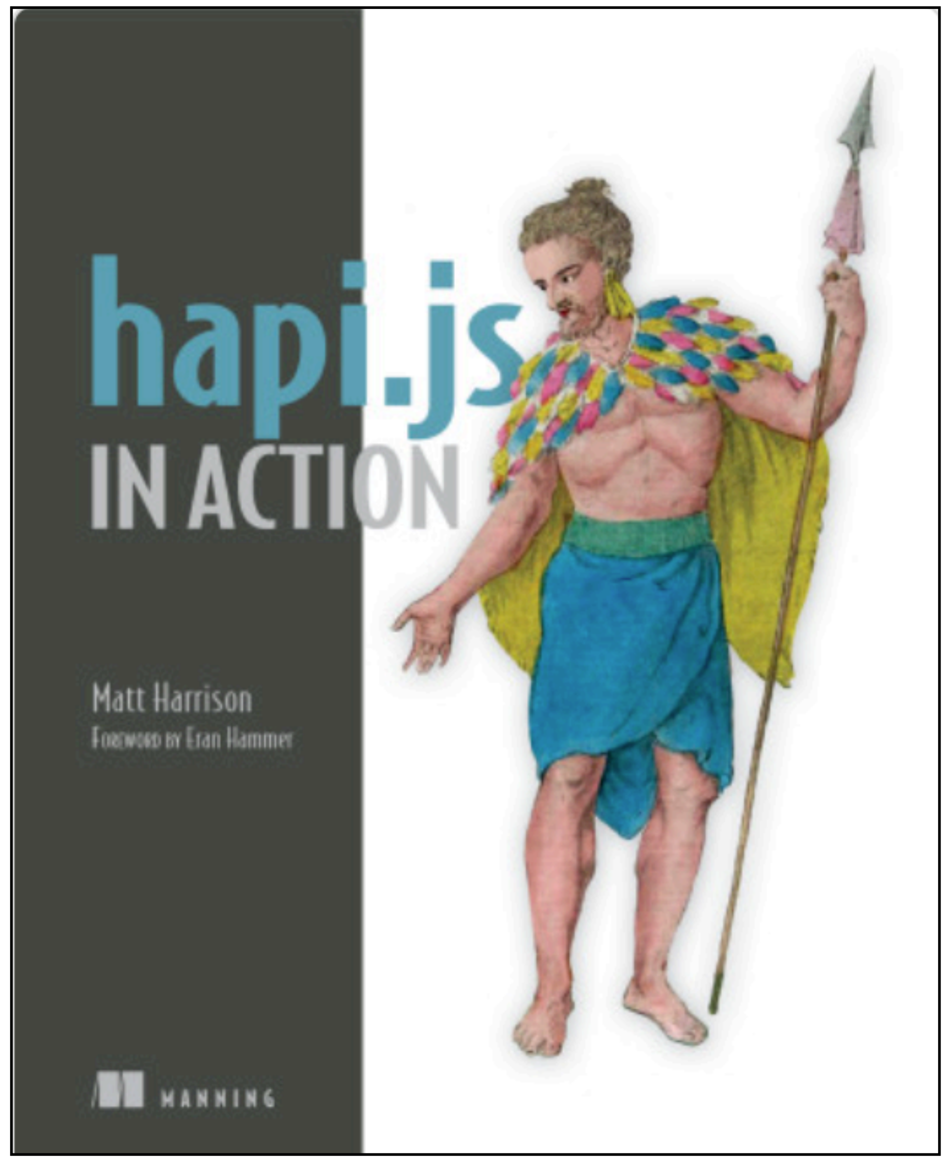
OPENAPI

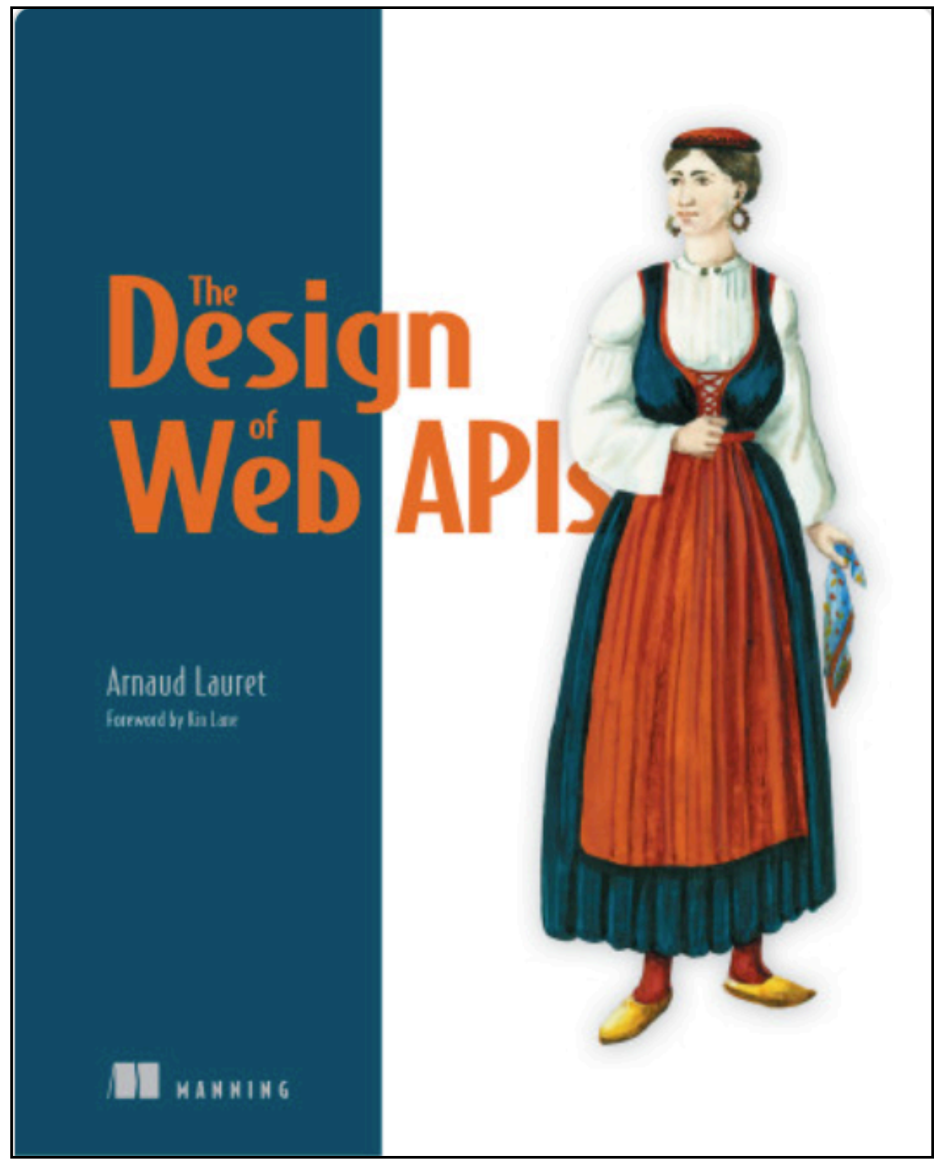
Robo 3T

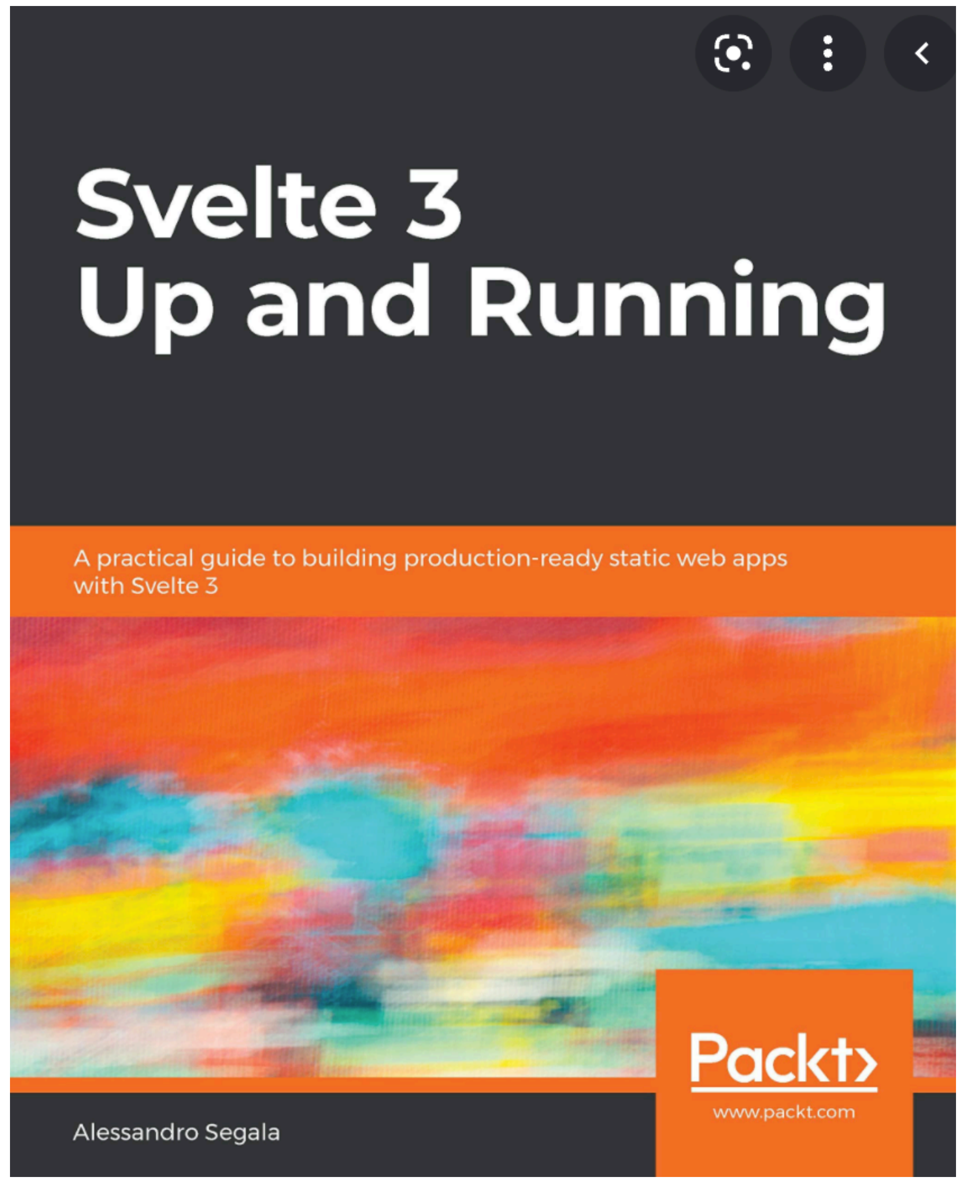
Frappe

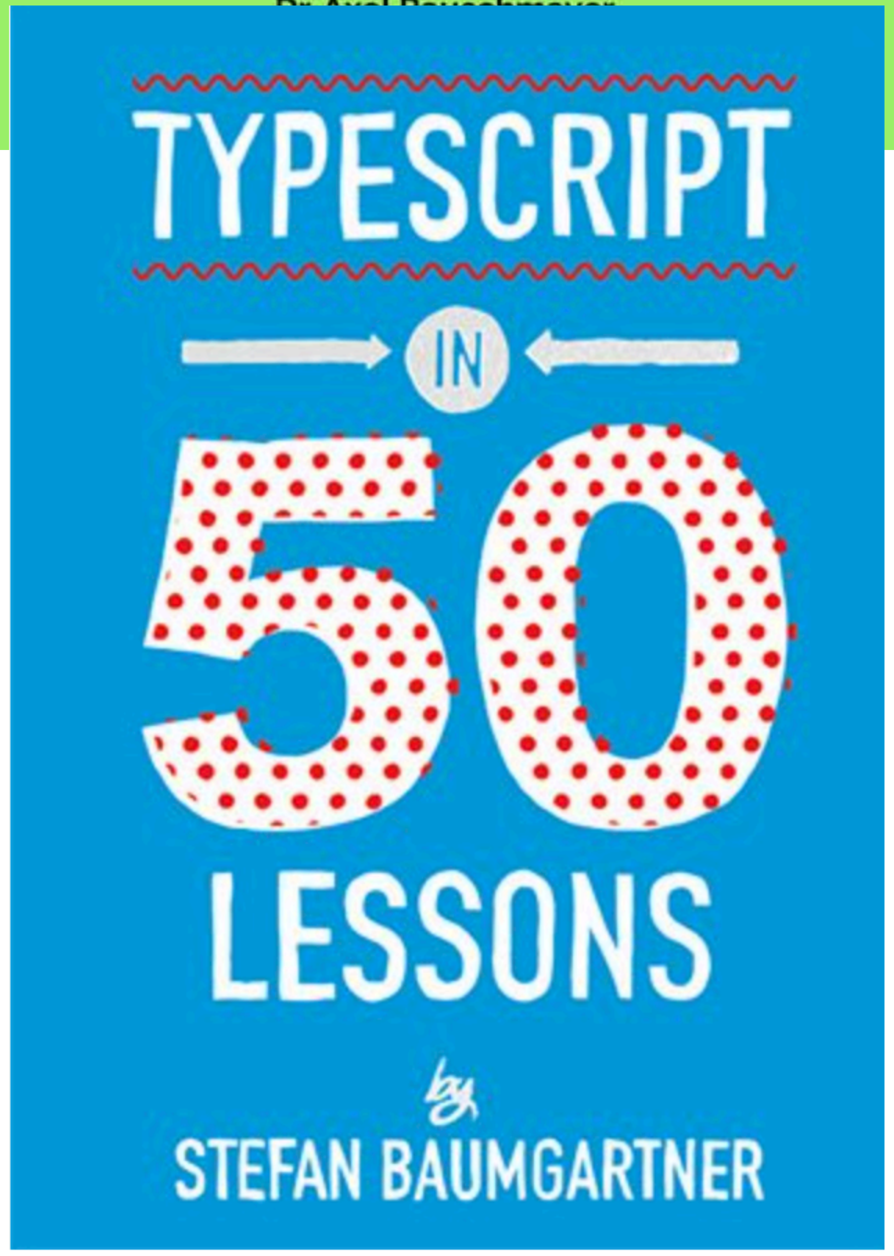
Heroku Dev Center



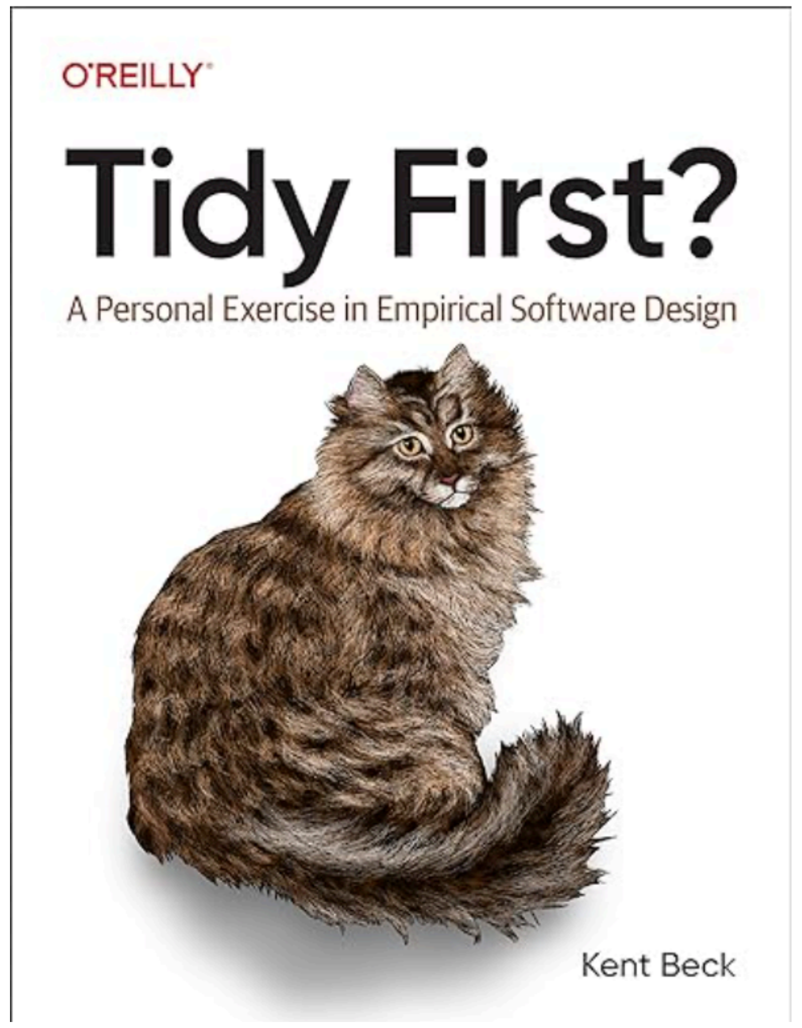


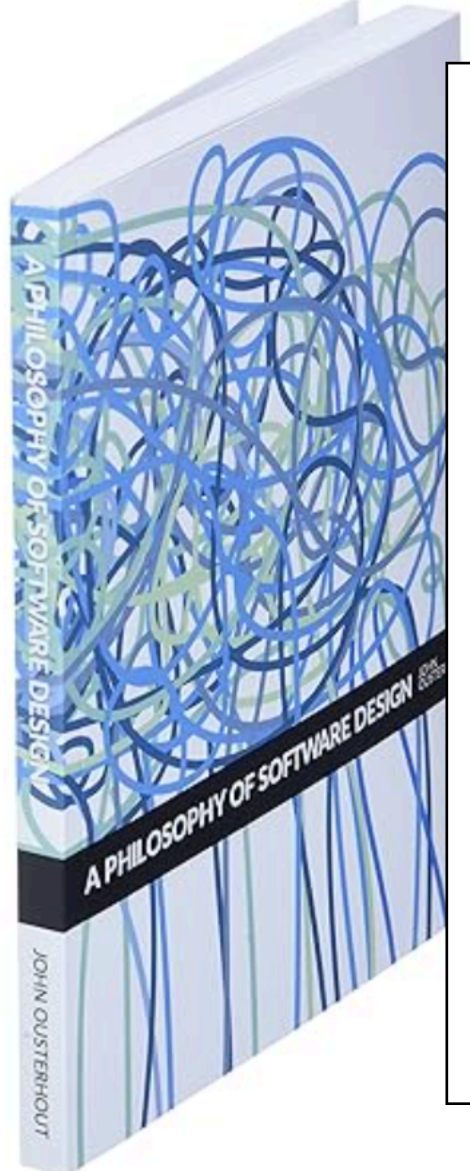


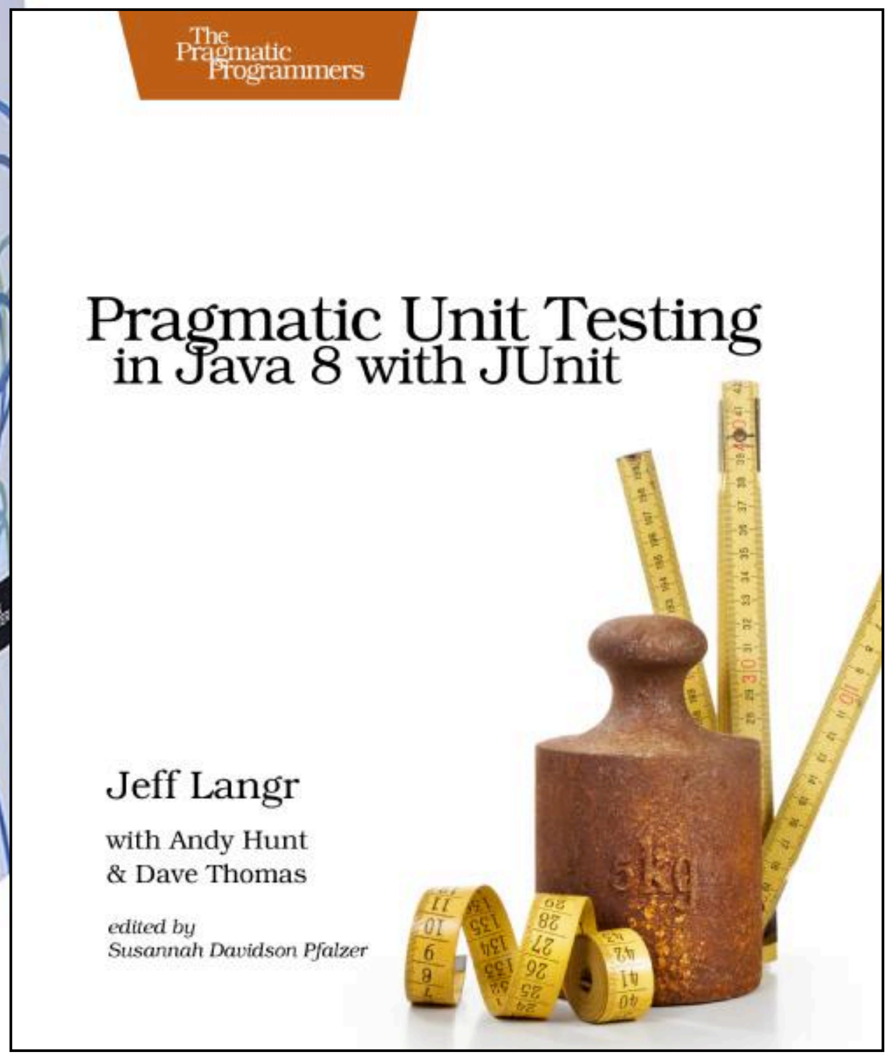




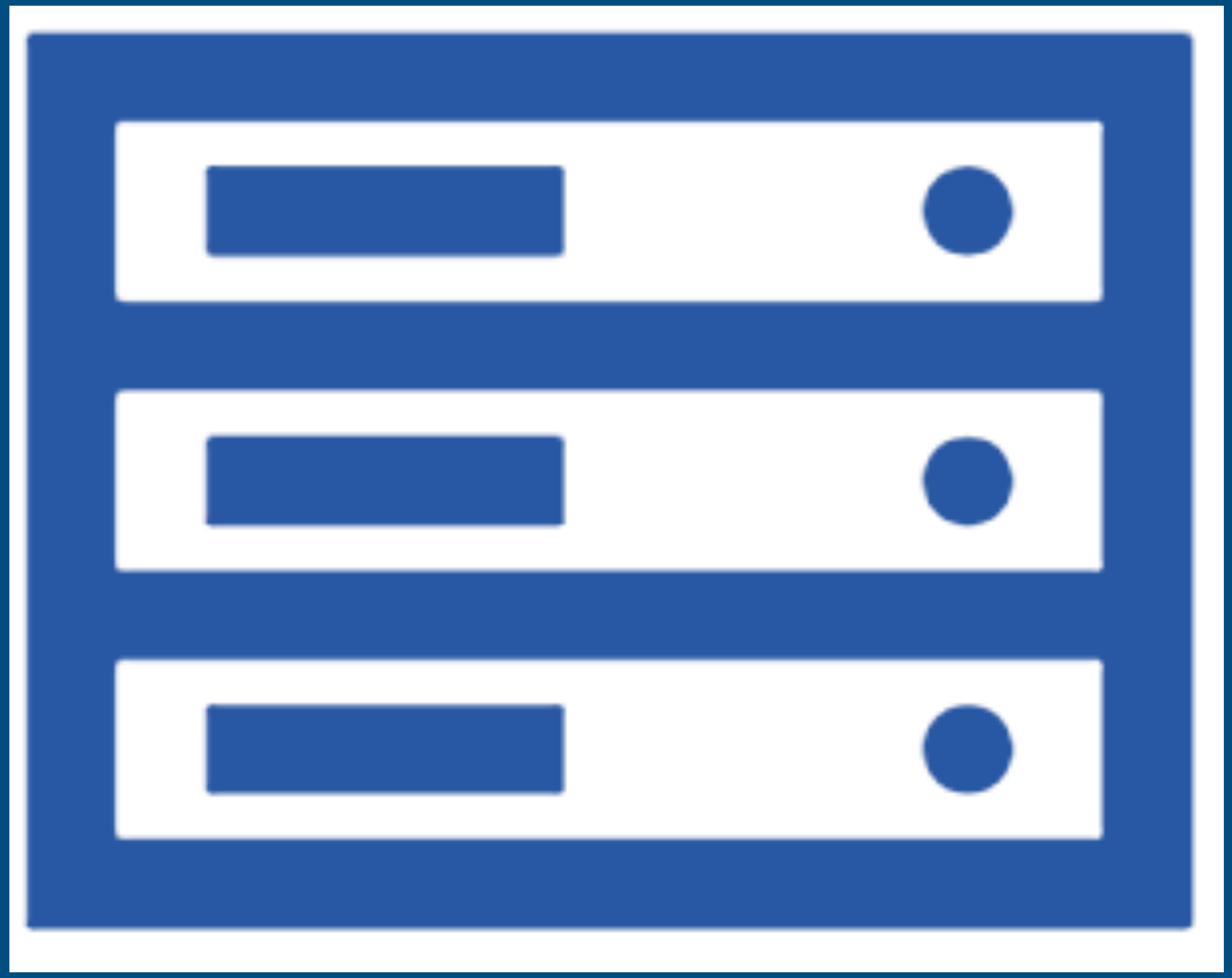
- https://exploringjs.com/impatient-js/toc.html
- https://nodeweekly.com/
- https://javascriptweekly.com/
- https://frontendfoc.us/
- https://svelte.dev/blog







Course Overview



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